

q Find the mean of:

1. $[10, 20, 30, 40, 50]$

$$\text{mean} = \frac{10 + 20 + 30 + 40 + 50}{5}$$

$$\text{mean} = 30$$

2. $[5, 10, 15, 20]$

$$\text{mean} = \frac{5 + 10 + 15 + 20}{4} = \frac{50}{4}$$

$$\text{mean} = 12.5$$

3. $[12, 18, 24, 30, 36]$

$$\text{mean} = \frac{12 + 18 + 24 + 30 + 36}{5} = \frac{120}{5}$$

$$\text{mean} = 24$$

4. $[8, 16, 24, 32]$

$$\text{mean} = \frac{8 + 16 + 24 + 32}{4} = \frac{80}{4}$$

$$\text{mean} = 20$$

Q ~~Variance~~ Find the variance of:

1. [2, 4, 6]

→

mean:

$$\frac{2 + 4 + 6}{3} = 4$$

Squared difference:

$$(2-4)^2 + (4-4)^2 + (6-4)^2 = 8$$

$$\text{Sum} = 8$$

$$\text{Variance} = \frac{8}{3} = 2.67$$

$$\text{variance} = 2.67$$

2. [10, 20, 30]

→

mean:

$$\frac{10 + 20 + 30}{3} = 20$$

Squared difference:

$$(10-20)^2 + (20-20)^2 + (30-20)^2 = 200$$

$$\text{Sum} = 200$$

$$\text{variance} = \frac{200}{3} = 66.67$$

$$\text{variance} = 66.67$$

• Quartile Practice question

Q Find Q_1 , Q_2 , and Q_3 for:

1. $[10, 20, 30, 40, 50, 60, 70, 80]$
→

$$Q_2 \text{ (median)} = \frac{40 + 50}{2} = 45$$

Lower half: $[10, 20, 30, 40]$:

$$Q_1 = \frac{20 + 30}{2} = 25$$

Upper half: $[50, 60, 70, 80]$:

$$Q_3 = \frac{60 + 70}{2} = 65$$

$$Q_1 = 25, \quad Q_2 = 45, \quad Q_3 = 65$$

2. $[5, 10, 15, 20, 25, 30, 35, 40]$
→

Q_2 (median):

$$Q_2 = \frac{20 + 25}{2} = 22.5$$

Lower half: $[5, 10, 15, 20]$

$$Q_1 = \frac{10 + 15}{2} = 12.5$$

Upper half: $[25, 30, 35, 40]$.

$$Q_3 = \frac{30 + 35}{2} = 32.5$$

$$Q_1 = 12.5, \quad Q_2 = 22.5, \quad Q_3 = 32.5$$

5. $[25, 35, 45, 55, 65]$

$$\text{mean} = \frac{25 + 35 + 45 + 55 + 65}{5} = \frac{225}{5}$$

$$\text{mean} = 45$$

Q Find the median of :

1. $[10, 20, 30, 40, 50]$

→

$$\text{median} = 30$$

2. $[5, 10, 15, 20]$

→

$$\text{median} = \frac{10 + 15}{2} = 12.5$$

$$\text{median} = 12.5$$

3. $[12, 18, 24, 30, 36]$

→

$$\text{median} = 24$$

4. $[7, 14, 21, 28]$

→

$$\text{middle values} = 14 \text{ and } 21$$

$$\text{median} = \frac{14 + 21}{2} = 17.5$$

$$\text{median} = 17.5$$

5. [11, 22, 33, 44, 55]

→

median = 33

Q Find the mode of:

1. [10, 20, 20, 30, 40]

→ 20 appears 2 times

mode = 20

2. [5, 5, 10, 15, 20]

→

5 appears 2 times

mode = 5

3. [7, 8, 8, 9, 10]

→

8 appears 2 times

mode = 8

4. [25, 25, 30, 35, 40]

→

25 appears 2 times

mode = 25

5. [1, 2, 2, 3, 4]

→

2 appears 2 times

mode = 2

3. [5, 5, 5]

→

mean :

$$\frac{5 + 5 + 5}{3} = \frac{15}{3} = 5$$

squared difference:

$$(5-5)^2 + (5-5)^2 + (5-5)^2 = 0$$

$$\text{variance} = 0$$

4. [1, 3, 5]

→

mean:

$$\frac{1 + 3 + 5}{3} = 3$$

squared differences:

$$(1-3)^2 + (3-3)^2 + (5-3)^2 = 8$$

$$\text{Sum} = 8$$

$$\text{variance} = \frac{8}{3} = 2.67$$

$$\text{variance} = 2.67$$

5. [4, 8, 12]

mean:

$$\frac{4 + 8 + 12}{3} = 8$$

Squared differences:

$$(4-8)^2 + (8-8)^2 + (12-8)^2 = 32$$

$$\text{Sum} = 32$$

$$\text{Variance} = \frac{32}{3} = 10.67$$

$$\text{Variance} = 10.67$$

Q Find the standard deviation of:

1. $[2, 4, 6]$

→

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

$$\text{Variance} = 2.67$$

$$\text{Standard deviation} = \sqrt{2.67} \approx 1.63$$

$$\text{Standard deviation} = 1.63$$

2. $[10, 20, 30]$

→

$$\text{variance} = 66.67$$

$$\text{SD} = \sqrt{66.67} \approx 8.16$$

$$\text{standard Deviation} = 8.16$$

3. $[5, 5, 5]$

→

$$\text{variance} = 0$$

$$SD = \sqrt{0} \approx 0$$

$$\text{standard Deviation} = 0$$

4. $[4, 3, 5]$ $[4, 8, 12]$

→

$$\text{Variance} = 10.67$$

$$SD = \sqrt{10.67} \approx 3.27$$

$$\text{standard Deviation} = 3.27$$

5. $[1, 3, 5]$

→

$$\text{variance} = 2.67$$

$$SD = \sqrt{2.67} \approx 1.63$$

$$\text{Standard Deviation} = 1.63$$

3. $[2, 4, 6, 8, 10, 12, 14, 16]$

→

Q_2 (median):

$$Q_2 = \frac{8 + 10}{2} = 9$$

lower half: $[2, 4, 6, 8]$

$$Q_1 = \frac{4 + 6}{2} = 5$$

upper half: $[10, 12, 14, 16]$

$$Q_3 = \frac{12 + 14}{2} = 13$$

$$Q_1 = 5, Q_2 = 9, Q_3 = 13$$

4. $[1, 2, 3, 4, 5, 6, 7, 8]$

→

$$Q_2 = \frac{4 + 5}{2} = 4.5$$

lower half: $[1, 2, 3, 4]$

$$Q_1 = \frac{2 + 3}{2} = 2.5$$

upper half: $[5, 6, 7, 8]$

$$Q_3 = \frac{6 + 7}{2} = 6.5$$

$$Q_1 = 2.5, Q_2 = 4.5, Q_3 = 6.5$$

5. [12, 24, 36, 48, 60, 72, 84, 96]

→

Q_2 (median):

$$Q_2 = \frac{48 + 60}{2} = 54$$

Lower half: [12, 24, 36, 48]

$$Q_1 = \frac{24 + 36}{2} = 30$$

Upper half: [60, 72, 84, 96]

$$Q_3 = \frac{72 + 84}{2} = 78$$

$$Q_1 = 30, Q_2 = 54, Q_3 = 78$$

⊕ Percentiles questions

1. Find the 25th Percentile of:
[10, 20, 30, 40, 50]

$$\text{Position} = \frac{P}{100} \times (n+1)$$

$$n = 5$$

$$\text{Position} = \frac{25}{100} \times (5+1) = 1.5$$

The 1.5th Position lies halfway between 10 and 20
 $10 + 0.5(20-10) = 15$

$$25^{\text{th}} \text{ Percentile} = 15$$

2. Find the 50th percentile of [5, 10, 15, 20, 25]

$$\text{Position: } \frac{50}{100} \times (5+1) = 3$$

The third value is 15

$$50^{\text{th}} \text{ Percentile} = 15$$

Find the 75th Percentile of [2, 4, 6, 8, 10]

$$\text{Position: } \frac{75}{100} \times (5+1) = 4.5$$

Between the 4th value (8) and 5th value (10)
 $8 + 0.5(10-8) = 9$

$$75^{\text{th}} \text{ Percentile} = 9$$

4. Find the 90th Percentile of [10, 20, 30, 40, 50]
→

$$\text{Position: } \frac{90}{100} \times (5+1) = 5.4$$

This lies beyond last values, so we take highest value.

90th Percentile = 50

5. Find the 60th Percentile of [12, 24, 36, 48, 60]
→

$$\text{Position: } \frac{60}{100} \times (5+1) = 3.6$$

Between the 3rd value (36) and 4th value (48)
~~36~~ $36 + 0.6(48 - 36)$

$$36 + 7.2 = 43.2$$

60th Percentile = 43.2